

Online Resource 1

Article title. How Much Ontology Does a Manufacturing Knowledge Graph Need? A Construction Ladder, a Cost Model, and a Decision Framework for Process Engineers

Journal. Journal of Intelligent Manufacturing

Authors. Michael J. McNary, Jian Liu, Donald C. Wunsch II

Corresponding author. Michael J. McNary, Department of Engineering Management and Systems Engineering, Missouri University of Science and Technology, Rolla, Missouri, USA; mmf22@mst.edu

Caption. Online Resource 1: Study-level appraisal of the primary comparative studies against the six adequacy criteria of §5.1 in the main text. ● satisfied; ◐ partly; ○ not; n/a not applicable. C1 relevant adjacent comparator; C2 evaluation independence; C3 non-focal components controlled; C4 resolution stated and nulls bounded; C5 industrial corpus and real task; C6 cost reported. Table 3 in the main text summarizes these rows by decision.

Study	Decision	Status	Corpus	C1	C2	C3	C4	C5	C6	Principal limitation for the ladder question
Han et al. (2025)	D1	Preprint (arXiv:2502.11371, version 3, March 2026)	Open-domain QA benchmarks	◐	●	●	○	○	◐	Dense-only plain-retrieval comparator; point accuracies only; costs reported as time, storage, and tokens rather than money; not industrial
Zeng et al. (2025)	D1	Preprint (arXiv)	Standard GraphRAG benchmarks, corrected	●	●	●	○	○	○	Evaluation-method study; not industrial
Sarmah et al. (2024)	D1	Peer reviewed (ICAIF 2024)	Financial filings	●	●	◐	○	◐	○	Hybrid vs. single channel; retrieval varies by design
Hoang et al. (2026)	D1	Peer reviewed (IJPE)	CNC process planning, 280 industry questions	◐	◐	○	○	●	○	Single system vs. authors' baselines; retrieval and construction vary together
Gao et al. (2026)	D1	Peer reviewed (ESWA)	Machine tool fault data	◐	◐	○	○	●	○	Single system; end-to-end accuracy only
da Cruz et al. (2025)	D1, D2	Preprint (arXiv)	One grant document, 20 questions	●	○	◐	○	○	○	Twenty items; chunk attachment varies across arms
Sharma et al. (2025), OG-RAG	D2	Peer reviewed (EMNLP 2025)	Agriculture, news	◐	◐	○	○	○	○	Comparator is plain retrieval (L1), not an open-schema graph; retrieval mechanism changes with representation
Zhang and Soh (2024), EDC	D2	Peer reviewed (EMNLP 2024)	Public KG benchmarks	●	●	n/a	○	○	○	Extraction quality, not downstream answers
Mihindukula sooriya et al. (2023), Text2KGBenc	D2	Peer reviewed (ISWC 2023)	Public text-to-KG benchmarks	○	●	n/a	○	○	○	Conformance and fidelity of triples, not answers

Study	Decision	Status	Corpus	C1	C2	C3	C4	C5	C6	Principal limitation for the ladder question
h										
van Cauter and Yakovets (2024)	D2	Peer reviewed (KaLLM 2024, ACL)	Maintenance short texts	○	●	n/a	○	●	○	Ontology-guided arm only
Zhou et al. (2026b)	D2	Peer reviewed (Applied Sciences)	Lifting-equipment failure reports	●	●	n/a	○	●	○	Extraction metrics; no open-schema graph at answer time
Okazaki et al. (2025), OGPAL	D2, D3	Preprint (arXiv)	Automotive FMEA worksheets	○	●	n/a	○	●	○	Ontology-guided arm only; link prediction outcome
Benson et al. (2024), My Ontologist	D3	Workshop (JOWO 2024, CEUR)	BFO definition-support tasks	○	○	n/a	○	○	○	Conformance achievability, not task outcome
Fathallah et al. (2024, 2026), NeOn-GPT	D4	Peer reviewed (ESWC 2024; Semantic Web 2026)	Wine, life-science domains	●	○	●	○	○	●	Outcome measured on the signals fed back
Huang et al. (2024); Madaan et al. (2023)	D4	Peer reviewed (ICLR 2024; NeurIPS 2023)	Reasoning benchmarks	●	n/a	●	○	○	○	General self-correction, not ontologies

Search audit: final targeted search for D3 and D5

Purpose. Decisions D3 (grounding in BFO or IOF Core) and D5 (grounding crossed with iteration) are graded Absent in the main text. This section documents the targeted search run to confirm that no qualifying study was located. A qualifying study for D3 builds ontologies from the same document corpus at two or more grounding depths (ungrounded, BFO, IOF Core, or comparable upper or mid-level ontology), populates graphs from documents the ontologies were not built from, and reports a downstream task outcome. A qualifying study for D5 crosses grounding depth with iterative refinement depth in one design.

Search date. 30 September 2026.

Sources. Scopus; IEEE Xplore; ACM Digital Library; Google Scholar. Google Scholar does not report a reproducible hit count, so the figures given are the records screened from the leading result pages for each string; 206 records were screened across the four strings, of which 140 were distinct.

Search strings. The same four strings were run in each source, adapted to its syntax (Scopus TITLE-ABS-KEY, IEEE Xplore and ACM DL full-text, Google Scholar plain). S1 and S2 target D3; S3 targets D5; S4 is a broad control.

- S1: (“Basic Formal Ontology” OR BFO OR “Industrial Ontologies Foundry” OR “IOF Core” OR “upper ontology” OR “top-level ontology”) AND (“large language model” OR LLM OR GPT) AND (ontology) AND (generat* OR construct* OR learn*)
- S2: (“ontology” AND (“knowledge graph” OR “retrieval-augmented”)) AND (“BFO” OR “upper ontology” OR “top-level ontology” OR “mid-level ontology”) AND (evaluat* OR accuracy OR “question answering” OR downstream)

- S3: (“ontology” AND (“large language model” OR LLM)) AND (iterat* OR refine* OR “self-correct” OR *feedback*) AND (*ground* OR scaffold* OR “upper ontology” OR BFO)
- S4: (“ontology” AND (“large language model” OR LLM) AND manufactur) AND (*compar* OR evaluat*)

Results.

Source	String	Hits	Screened (title/abstract)	Read in full	Qualifying for D3	Qualifying for D5
Scopus	S1	11	9	1	0	0
Scopus	S2	2	1	0	0	0
Scopus	S3	40	35	1	0	0
Scopus	S4	56	46	2	0	0
IEEE Xplore	S1	2	2	0	0	0
IEEE Xplore	S2	1	1	0	0	0
IEEE Xplore	S3	12	12	0	0	0
IEEE Xplore	S4	13	13	0	0	0
ACM Digital Library	S1	34	11	0	0	0
ACM Digital Library	S2	62	40	0	0	0
ACM Digital Library	S3	1,000 (export cap; full-text search)	657	0	0	0
ACM Digital Library	S4	775	148	0	0	0
Google Scholar	S1	46 (first pages screened)	46	3	0	0
Google Scholar	S2	46 (first pages screened)	46	0	0	0
Google Scholar	S3	55 (first pages screened)	55	0	0	0
Google Scholar	S4	59 (first pages screened)	59	0	0	0

Note on the ACM Digital Library counts. The ACM search was run over full text (“Anywhere”), which returns proceedings front-matter records and papers that match the terms only in passing. Hit counts are as returned; the S3 export was capped at 1,000 records by the platform. “Screened” counts papers after removal of proceedings front-matter records. Screening proceeded in two steps: a keyword filter on title and abstract (ontology, knowledge graph, or taxonomy, together with language model, LLM, or GPT), which retained 7, 6, 197, and 26 records for S1 to S4, followed by title and abstract review of those records.

Studies read in full. Hit counts exclude proceedings front-matter records. Screening was by title and abstract against the qualifying definitions above. The seven studies below came closest or bore most directly on the manuscript and were read in full; this is the total in the “Read in full” column of the results table.

- Bjarsch et al. (2026), LLM benchmark for BFO and CCO subclass decisions (Scopus, S1); failed: no downstream outcome (D3 condition 3). Cited in §5.4 as conformance evidence.

- Cotti et al. (2026), OntoLogX, ontology-guided extraction with iterative correction (Scopus, S3); failed: single grounding condition, iteration not crossed with grounding (D5).
- Yang et al. (2026), ontology-based knowledge graph QA for equipment maintenance (Scopus, S4); single system, single grounding condition; cited in §5.2 as an industrial demonstration.
- Oyekan et al. (2025), decision-making guidance on LLMs, ontologies, and knowledge graphs for automation (Scopus, S4); not a comparative study; cited in §2.1 as a related review.
- Gautam (2025), master's thesis on ontologies and data structures in RAG over manufacturing simulation catalog data (Google Scholar, S1); failed: compares data representation formats and retrieval architectures over structured catalog records, not schema presence or grounding depth over a fixed document corpus; 50 items scored by embedding similarity and human judgment (D1, D2). Cited in §5.2 as grey literature consistent in direction.
- Yang et al. (2025), SSRN preprint, hand-built automotive assembly ontology on BFO and IOF Core with LLM text-to-Cypher retrieval (Google Scholar, S1); failed: ontology not model-built, single grounding condition, ablation varies prompt components rather than grounding (D3). Cited in §3.2 as an IOF Core application.
- Bhattacharya et al. (2026), ESWC 2026, factor study of LLM ontology population over three lightweight ontologies and four LLMs (Google Scholar, S1); failed: no upper-ontology grounding condition (the snippet's upper-ontology mention is background), no downstream task (D3). Cited in §5.3 for the domain-familiarity finding.

Other near misses, excluded at title and abstract. The studies below were excluded on the basis of title and abstract. They are listed because each came close to one of the qualifying definitions, so that the reason for exclusion is on record. Where one of these is cited in the manuscript, the citation rests on what its abstract reports.

- Schmidt et al. (2026), systematic literature review of neuro-symbolic and LLM-based knowledge graph construction in manufacturing (IEEE Xplore, S4); not a comparative study; cited in §2.1 as a related review.
- Lopes et al. (2024), classification of domain entities into top-level ontology concepts (Scopus, S1); failed: no downstream outcome (D3 condition 3).
- Rodrigues et al. (2023), classification of domain terms under upper ontologies with ChatGPT (Scopus, S4); failed: no downstream outcome (D3 condition 3).
- Procko et al. (2023), BFO-compliant definition generation with GPT (Scopus, S1); failed: no downstream outcome (D3 condition 3).
- Abolhasani et al. (2025), OntoKGen, iterative ontology extraction (Scopus, S3); failed: single grounding condition (D5).
- Liu et al. (2026), comparison of LLM extraction strategies for a casting ontology (Scopus, S4); failed: compares extractors, not schema presence or grounding depth (D2, D3).
- Guryanov et al. (2025), LLM-assisted OWL ontology extraction from ISO documentation (IEEE Xplore, S4); failed: single construction condition, no grounding comparison, extraction outcome only.

- Xie et al. (2026), LLM-driven multimodal industrial KG construction with iterative prompt optimization (IEEE Xplore, S4); failed: single grounding condition, iteration not crossed with grounding (D5).
- Zhao et al. (2025), ontology-guided KG retrieval versus a hybrid BM25-plus-semantic baseline on multi-hop store-fulfillment queries (ACM DL, S3); not a document-corpus study and no schema-depth comparison; cited in §5.2 as an example of the hybrid comparator form.
- Zaitoun et al. (2023), automated ontology evaluation of coverage and correctness against a domain corpus (ACM DL, S3); an evaluation method, not a comparative study; cited in §8.
- Sun et al. (2024), whether LLMs replace taxonomies (ACM DL, S3); not a construction study; cited in §3.1.
- Carta et al. (2024), zero-shot automated schema inference for KG building (ACM DL, S3); failed: no comparison against a predefined-schema arm on the same corpus with a downstream outcome (D2).
- Bombieri et al. (2026), memorization of ontology identifiers by LLMs (ACM DL, S1); not construction, no grounding comparison.
- Hou et al. (2025), classification of cultural-heritage terms into an upper-level ontology (ACM DL, S1); failed: no downstream outcome (D3 condition 3).
- Yhdego and Wang (2025), automated ontology generation for zero-shot defect identification, with a comparative study against a manually generated ontology (Google Scholar, S4); failed: compares construction route, not schema presence or grounding depth (D2, D3). Cited in §5.3.
- Wang et al. (2024), ontology-integrated tuning of an LLM for intelligent maintenance (Google Scholar, S4); failed: compares options for incorporating an ontology into a fine-tuned model, not schema presence (D2). Cited in §5.3.
- Babaei Giglou et al. (2026), controlled study of LLM scale for ontology learning (Google Scholar, S2); failed: varies model scale, not grounding depth; no downstream task (D3).
- Bengtsson et al. (2024), ontology plus LLM for recurring machine failures in free-text fields (Google Scholar, S4); failed: single system, no comparator without the ontology (D2).
- Robasky et al. (2026), MCBO, an IOF-anchored bioprocessing application ontology (Google Scholar, S3); failed: hand-built, single grounding condition, no downstream outcome (D3).
- Lisena et al. (2026), pitfalls in AI-generated ontologies with debugging strategies (Google Scholar, S3); failed: quality-assurance method, not a grounding or schema comparison.
- Bottino and Alcazár (2026), scoping review of KG construction from text with LLMs (Google Scholar, S2); a review, not a comparative study; overlaps the reviews positioned in §2.1.

Mapping of located studies to decisions. The Decision column of the table above records which decision each study was appraised against. A study bears on a decision when it compares, or provides evidence about, the two rungs that decision separates.